

Screen A: detailed design

Two prompts and one authorization question, and Claude Code produced four detailed-design documents, a rendered mockup published as a private artifact, four decisions — and a change to the harness that produced them.

Presented alongside `ScreenA_DD_edited.mp4` (4 min 00 s). Step numbers on the slides match the captions in the video.
Recorded 2026-09-18 · Claude Code (Opus 5) in the JetBrains Rider terminal, auto mode on.

`DD-001-production-order-create-edit.md`

`DD-001-API-production-orders.md`

`DD-001-FN-production-order-service.md`

`DD-001-SPD-...-create-edit.md`

`DD-001-screen-a-mockup.html`

`decisions.md · status.md · evidence.md`

Sixteen minutes of recording, ten and a half of agent work

The human wrote two prompts and answered one question. Everything else was the agent working through the harness.

2

prompts typed

one of them a correction

1

authorization asked

answered in 4 seconds

10 m 23 s

agent working time

7 m 27 s + 2 m 56 s

1,314

lines of design

across four documents

The human

- Reviewed the database design, then asked for the detailed design.
- Cleared the one permission the plan had deferred, before any work depended on it.
- Noticed that two of the four DD templates had produced no document — and said so in one line.

The agent, guided by the harness

- Read the plan's authorization table and stopped to ask, rather than discovering the gate halfway through.
- Read the running app's own CSS before designing a mockup for it.
- Fed a gap found while writing the API contract back up into the basic design.
- Turned its own mistake into a memory, then into a proposal to change the shared templates.

12 steps, one video

Each step is a clip in the edited video. **Amber: the human acts.** **Teal: the agent works and produces output.**

STEP 0100:00 Prompt 1 DB design reviewed, start the DD	STEP 0200:12 Reads first Skills, templates, and the app's own code	STEP 0300:32 Authorization May the mockup be published?	STEP 0400:50 DD-001 written The detailed design
STEP 0501:03 DD-001-API written Split out for Screen B to reuse	STEP 0601:16 Mockup approach Reuses the app's own palette	STEP 0701:38 Mockup published Private artifact, 11 states	STEP 0801:51 Turn 1 summary DEC-021–DEC-024, BD-001 rev 4
STEP 0902:23 Review: DD-001 ...and the DD template folder	STEP 1002:37 Prompt 2 Two templates produced no document	STEP 1102:49 Memory, then the fix DD-001-FN, DD-001-SPD, rewiring	STEP 1203:47 Review the four docs ...plus the mockup

Times are positions in `ScreenA_DD_edited.mp4`. Long stretches of agent work are jump-cut to their result; nothing is sped up.

Review first, then hand the work on

THE PROMPT

I reviewed the DB design and it looks okay, so let's move on to detailed design for this work item

Before typing, the user scrolls `plan.md`: its **Authorization** table ("Publish rendered mockup as a private Artifact — ask at execution — not yet authorized — step 5"), its risks, and the sign-off recording revision 1 as approved.

OUTPUT

The prompt carries a review result, not just an instruction. Claude Code's suggested reply ("Move on to the detailed design") was cleared and replaced.

```
plan scope: because the login split is happening now, the implementation plan (revision 2) will need to change WI-001's docker compose files:
- a new required runtime password in .env.example, with no default;
- a second connection string;
- a script that grants the runtime login's permissions;
- updating the migration command in ai/project.md to run as the owner login.

That plan revision will come to you for review before any of it is built.

Next is step 5: the detailed design (DD-001) plus a rendered mockup. I'll ask before publishing the mockup.

* Sautéed for 1m 51s · done 11:12 AM

> i reviewed the DB design and it look okay so let move on to detailed design
  >> auto mode on (shift+tab to cycle)
```

Video 00:06. `plan.md` above, the prompt below.

Read the rules — and the running app

WHAT CLAUDE READ	WHAT IT GOT FROM IT
detailed-design and screen-design skills	The route through this step and its checklists
ai/templates/DD/*	The four DD templates and what each is for — the detail that matters in step 10
LoginPage.tsx, AuthContext.tsx, index.css	The app's real visual language — used an hour later to decide how the mockup should look
ADR-0001, the test projects	The layered backend contract and what test tooling already exists

OUTPUT

Claude does not start writing. It stops on a permission question instead — "Before writing: plan step 5 says publishing the rendered mockup as a private artifact needs your OK at that point, so I'm asking now so it doesn't hold things up later."



Video 00:18. Read-only commands, collapsed by Claude Code.

The gate was written down, so it was honoured

```
Next is step 5: the detailed design (DD-001) plus a rendered mockup. I'll ask before publishing the mockup.

* Sautéed for 1m 51s · done 11:12 AM

i reviewed the DB design and it look okay so let move on to detailed design for this work item

Read 3 files, listed 1 directory, ran 3 shell commands

• Before writing: plan step 5 says publishing the rendered mockup as a private artifact needs your OK at that point, so I'm asking now so it doesn't hold things up later.

· Photosynthesizing... (1m 12s · ↓ 6.6k tokens)

> |

>> auto mode on (shift+tab to cycle) · esc to interrupt · ← for agents
```

Video 00:41. One question, two real options, answered in about four seconds.

WHAT WAS ASKED

"DD-001 includes a rendered mockup of Screen A covering create, edit, locked-fields, validation-error, success and discard-dialog states. Can I publish it as a private claude.ai artifact (only you can see it unless you share it)?" — **Yes, publish privately** · Local HTML file only · Type something

WHY IT MATTERS

Nobody re-stated the rule in the prompt. It was a row in `plan.md`'s authorization table — *ask at execution* — written when the plan was drafted. The agent read it, honoured it, and cleared it **before** starting the work that depended on it, not halfway through.

The design, and the contract split out from it

STEP 4 — DD-001-PRODUCTION-ORDER-CREATE-EDIT.MD

609 lines: modules, screen states, processing flow, persistence, exception handling, observability and test viewpoints. Its tables map every BD rule to a concrete place in the code — `ProductionOrder.ValidateQuantity` for V-02/V-05, `IPlantClock.Today` for V-04, request-vs-entity `xmin` for V-08.

```
1
2
3 # Production Order Create/Edit - Detailed Design Document (詳細設計書)
4
5 DD-001 - Implements BD-001 (SCR-001), requirements REQ-010-REQ-019.
6
7 ## Document control (改訂履歴)
8
9 | Field | Value |
10 | --- | --- |
... +599 lines
```

Video 00:56. DD-001 streams to the terminal.

STEP 5 — DD-001-API-PRODUCTION-ORDERS.MD

292 lines. Split into its own document **because Screen B will reuse `/api/products`** — a decision about the next work item, made while writing this one.

```
1
2
3 # Production Orders API - API Specification Design (API仕様設計)
4
5 DD-001-API - supports DD-001, requirements REQ-010-REQ-018.
6
7 ## Document control (改訂履歴)
8
9 | Field | Value |
10 | --- | --- |
... +282 lines
```

Video 01:09. The API specification.

DESIGN FEEDS BACK UPSTREAM

Writing the contract exposed a missing quantity upper bound. It became **DEC-024** and **BD-001 revision 4** — not a silent choice in the code.

It looks like the product, because it read the product

STEP 6 — BEFORE BUILDING

"The DD documents are written. Before building the mockup I need to load the artifact design guidance." → `Skill(artifact-design)`

...AND THE VISUAL DECISION

"I'll build the mockup using the app's existing light-only Tailwind gray palette from `LoginPage.tsx` — gray-900 buttons, gray-300 borders, system font — while only the spec-sheet chrome around it adapts to the viewer's theme."

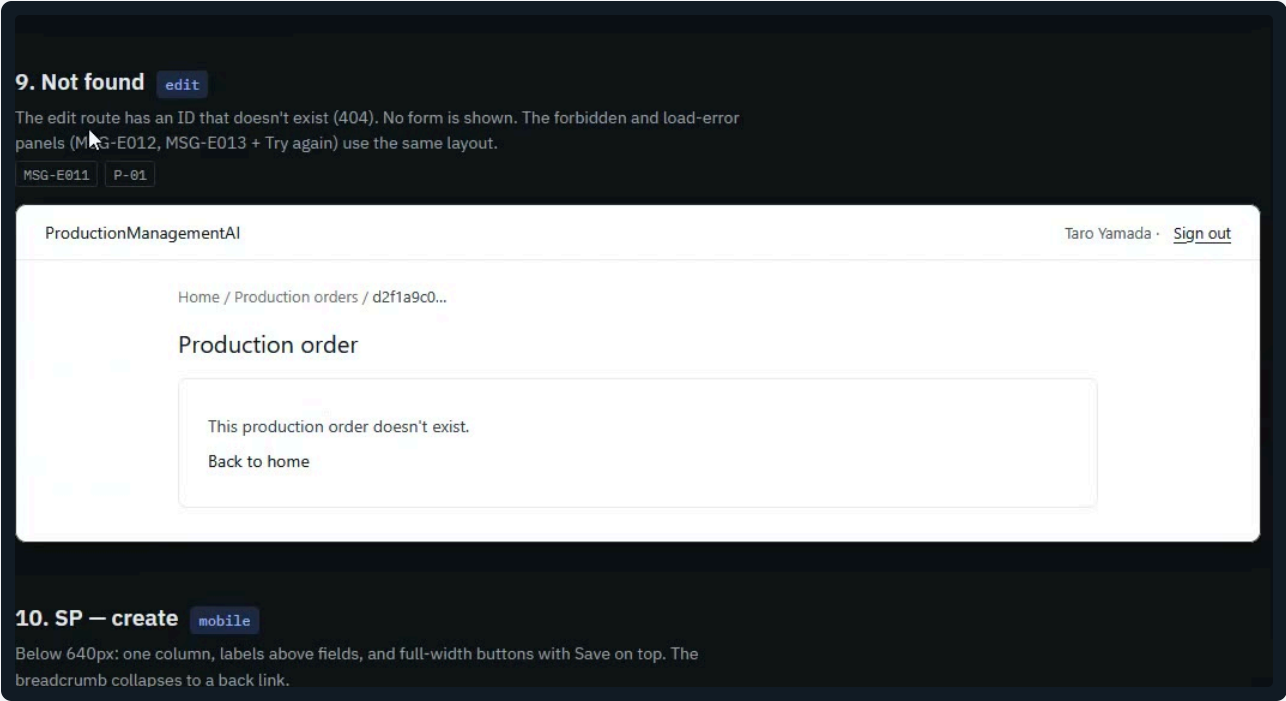
That is what reading `index.css` in step 2 was for.

```
• Publishing the mockup privately now.
• Artifact[D:\Work\AI\WMS\docs\en\020_detailed-design\mockups\DD-001-screen-a-mockup.html]
  ↳ Published 📄 https://claude.ai/artifact/FtG18d27Uj0xx5C9jx0Mq
• Published. Next: linking it from DD-001, fixing the order of DD-001's revision table, and updating status and evidence.
  ↳ Photosynthesizing... (on 54s · 4.40.1K tokens)
✓ Update installed · Restart to update
```

Video 01:44. Step 7: published to a private artifact URL.

`mockups/DD-001-screen-a-mockup.html` (292 lines) stays in the repository; the artifact is the rendered copy, linked from DD-001.

A design a non-developer can actually judge



Video 01:58. State 8 of 11 — the discard dialog, tagged REQ-019, DEC-018, items 17–19, E-07, E-09.

The artifact holds **11 static states**: create empty, product list open (30 items), validation errors, created → edit (Draft), in progress (locked), completed (terminal, overdue), stale save (409), discard dialog, not found, SP create, SP locked. Each carries the requirement and decision IDs it realises.

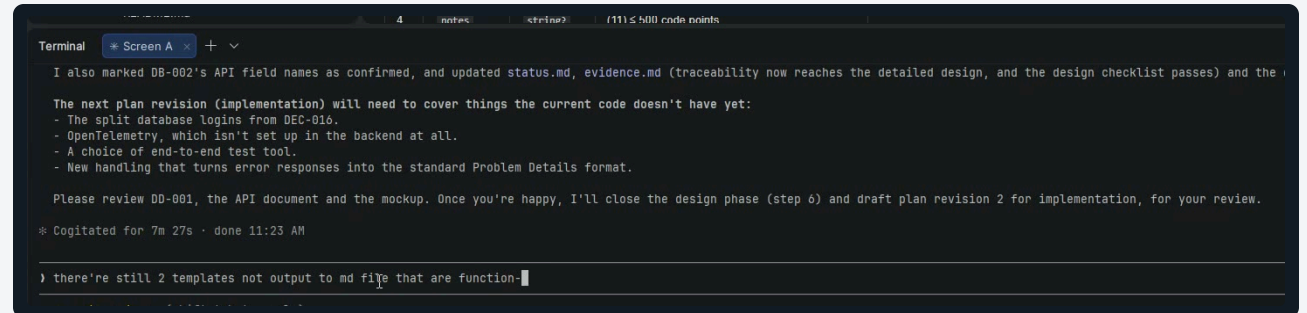
ID	DECIDED WHILE WRITING THE DESIGN
DEC-021	Browser-local date for early feedback; the server's plant date is authoritative
DEC-022	Native <code><select></code> ordered by SKU for 30 products
DEC-023	RFC 9457 error contract; <code>code</code> = message ID
DEC-024	Quantity upper bound 999,999,999 → BD-001 revision 4

The user almost accepted it

The user types a prompt accepting the work — *"The design looks okay, let's move on to the implementation"* — and **does not send it**. They read DD-001, then expand `ai/templates/DD/` in the project tree, where `api-design.md`, `function-design.md` and `screen-processing-design.md` sit side by side. Two of them produced no document.

THE PROMPT THAT WAS SENT INSTEAD

There are still 2 templates not output to an .md file: function-design and screen-processing-design



```
Terminal * Screen A x + v
I also marked DB-002's API field names as confirmed, and updated status.md, evidence.md (traceability now reaches the detailed design, and the design checklist passes) and the
The next plan revision (implementation) will need to cover things the current code doesn't have yet:
- The split database logins from DEC-016.
- OpenTelemetry, which isn't set up in the backend at all.
- A choice of end-to-end test tool.
- New handling that turns error responses into the standard Problem Details format.
Please review DD-001, the API document and the mockup. Once you're happy, I'll close the design phase (step 6) and draft plan revision 2 for implementation, for your review.
* Cogitated for 7m 27s · done 11:23 AM
} there're still 2 templates not output to md file that are function-
```

Video 02:43. The template folder open, and the correction being typed.

Fix the output, fix yourself, then propose the rule

"Understood. I left those two out as optional, but you want the full template family produced. **First I'll save that preference** so later design steps include all four."

SCOPE 1 — THIS AGENT

A memory, written first

Before producing anything, Claude wrote the preference down: *"in the WMS project, detailed design must output every DD template (main + api-design, function-design, screen-processing-design) as its own .md file"*. The order matters — the habit is fixed before the symptom.

SCOPE 2 — THIS WORK ITEM

The two missing documents

`DD-001-FN` (325 lines) and `DD-001-SPD` (212 lines). DD-001 was then rewritten to **point at them instead of repeating their content**, and the API document's cross-references, the docs index, `status.md` and `evidence.md` were updated to match.

SCOPE 3 — EVERYONE

A proposal, not a change

"The project's shared templates and the detailed-design skill still call the three companions optional. If you want that to be the rule for everyone, I can propose the change. **Changes to shared templates need your review before they take effect.**"

```
• Understood. I left those two out as optional, but you want the full template family produced. First I'll save that preference so later design steps include all four.

Searching for "### |###
1 Remembering in the WMS project, detailed design must output every DD template (main + api-design, function-design, screen-processing-design) as its own .md file
2 "### |###"

• Bootstrapping... (10s - 1.0k tokens)
  ↳ Tip: Working with HTML/CSS? Install the frontend-design plugin:
    /plugin install frontend-design@claude-plugins-official

✓ Update installed - Restart to update
```

Video 02:56. "Searching for 1 pattern, writing 1 memory..."

```
1 DD-001-FN - used by DD-001 and DD-001-API (and later by Screen B's list DD), requirements REQ-010-REQ-010.
2
3 ## Document control (### |###)
4
5 | Field | Value |
6 | --- | --- |
7 - 324 lines

• Now the screen-processing document. I'll reread the current flows (P-01-P-05) in DD-001 so I can jump to bottom [ctrl+End] ↓ anything.
```

Video 03:22. The second companion, then DD-001 rewired to both.

The design set stops repeating itself

DOCUMENT	LINES	ROLE
DD-001	485	The design; now delegates to the three companions
DD-001-API	292	The HTTP contract; reusable by Screen B
DD-001-FN	325	Function design, backend and frontend
DD-001-SPD	212	Screen processing and exception handling

OUTPUT

"I saved this as a preference in memory: for this project, detailed design always produces all four template documents." — *Worked for 2m 56s*

```
- API document: its processing steps now point to the matching sections in the two new documents.
- docs/en/020_detailed-design/README.md, status.md and the traceability matrix in evidence.md now include the new files.

I saved this as a preference in memory: for this project, detailed design always produces all four template documents.

The project's shared templates and the detailed-design skill still call the three companions optional. If you want that to be the rule for everyone, I can propose the change to those files. Changes to shared templates need your review before they take effect.

Next, as before I once you've reviewed the four detailed-design documents and the mockup, I'll close the design phase and draft plan revision 2 for implementation.

* Worked for 2m 56s · done 11:28 AM
```

Video 03:38. The turn 2 summary, ending on the offer to change the shared templates.

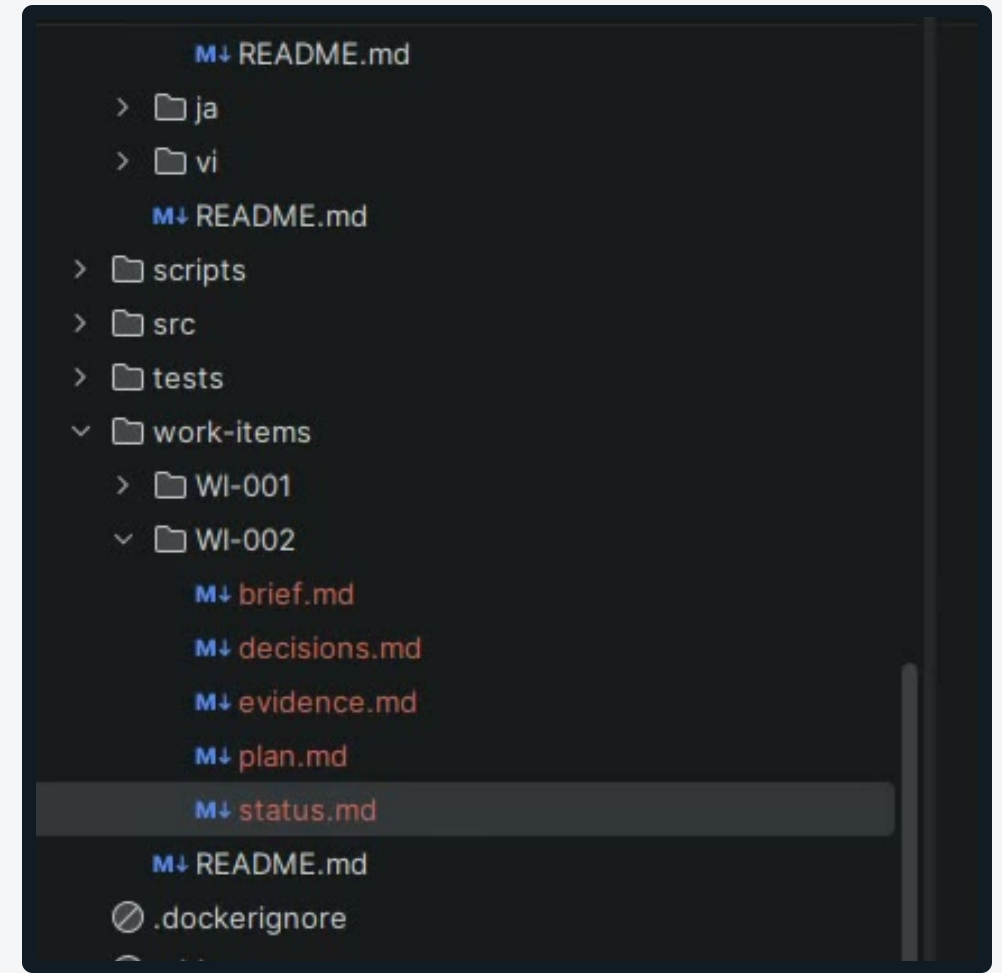
Four documents and a mockup, all traceable

The user reads each document in turn, ending on DD-001-SPD's exception-handling table — 401, 403, 404, 415, 400 with message IDs, 409 on a stale save, 422 on a rule violation, and 500 `MSG-E013` with no exception detail leaked.

Its closing section reads **"Unresolved decisions: None."** with the related decision IDs listed — DEC-007, DEC-008, DEC-009, DEC-018, DEC-020, DEC-021, DEC-023.

WHERE THE SESSION ENDS

Claude's suggested next prompt — "Update the templates and skill so all DD companions are always produced" — sits unsent in the input line. That became harness RFC 0001.



Video 03:53. 020_detailed-design, complete.

RFC 0001 — always produce all four DD documents

This is the one step in the Screen A series where using the harness exposed a defect **in the harness**, and the run produced a change proposal against it.

ROOT CAUSE — IN THE WORDING, NOT THE AGENT

The templates described the three companions as optional — "Use it when...", "None — this screen's design fits entirely in this file". Folding their content into the main DD was a *permitted reading* of the rule. The RFC records that, with the session as evidence.

WHAT THE RFC CHANGED

- `ai/templates/detailed-design.md` and the three `ai/templates/DD/*` templates
- `ai/templates/README.md`, `ai/skills/detailed-design/SKILL.md`
- `ai/evaluations/baseline-cases.md` — so the case is checked in future evaluations

THE BOUNDARY THAT WAS KEPT

The agent changed **its own** behaviour immediately, and **offered** the change to everyone's. Shared harness files are reviewed by their owner before they take effect — the same gate as the mockup in step 3, applied to the harness itself.

STATUS

Adopted. Recorded at `ai/improvements/0001-dd-companions-always-produced.md`.

One other walkthrough produced an RFC: the implementation step found that nothing told the harness to keep superseded plan revisions, which became **RFC 0002 — keep every plan revision, oldest first**.

Six things to notice

01

A written gate is honoured unprompted

`plan.md` said publishing needs an OK "at execution". Claude asked before starting, not halfway through — nobody restated the rule.

04

The human catches what the rule allowed

Two of four templates produced nothing. The user spotted it from the template folder and said so in one line.

02

It reads the app before designing for it

`LoginPage.tsx` and `index.css` were read in step 2 and used in step 6, so the mockup looks like the product.

05

Fix yourself before fixing the output

The preference went to memory *first*, then the missing documents were written. The habit is fixed before the symptom.

03

Design feeds back upstream

A missing quantity bound found while writing the API contract became DEC-024 and BD-001 revision 4, not a silent choice in code.

06

Shared rules stay with the human

Changing everyone's templates was offered as a proposal and became RFC 0001 after review — not applied unilaterally.

Design phase complete, implementation next

FILE	STATE AFTER THE SESSION
DD-001 + API + FN + SPD	Complete; 1,314 lines, cross-linked
DD-001-screen-a-mockup.html	In the repo; published as a private artifact
decisions.md	DEC-001 to DEC-024 decided
BD-001-...-create-edit.md	Revision 4
status.md	Green; step 6 and the DD review remain
evidence.md	Traceability reaches the detailed design
ai/improvements/0001-...md	Adopted

Sources: the session recording ([ScreenA_DD.mp4](#)), its transcript, and the committed files in [docs/en/020_detailed-design/](#) , [work-items/WI-002/](#) and [ai/improvements/](#) .

CARRIED INTO THE IMPLEMENTATION PLAN

- The split database logins from DEC-016.
- OpenTelemetry, which isn't set up in the backend at all.
- A choice of end-to-end test tool.
- Turning error responses into the standard Problem Details format.

NEXT IN THE SERIES

Implementation — code and tests at every level, and the step that produced the second harness RFC.